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Course Code: SBT-DF202

Course Title : Computer and Digital Forensics

Lab Title: Lab1 - Digital Forensics Case Handling,
Autopsy and Sleuth Kit Analysis

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Lab Objective

The objective of this laboratory exercise is to demonstrate the practical application of digital forensic investigation procedures in a controlled environment using the authorized forensic disk image Ch01InChap01.dd.

The investigation aims to:

1. Preserve the integrity of the original forensic evidence and document its identification, storage location, and cryptographic hashes.
2. Create and configure a forensic case in Autopsy and add the supplied disk image as the evidence source.
3. Identify and document deleted files within the forensic image.
4. Perform keyword searches to locate potentially relevant evidence within the raw forensic image.
5. Identify and locate the INCOME.XLS spreadsheet within the filesystem.
6. Examine the spreadsheet's filesystem metadata, including its metadata address, file size, timestamps, and allocated data units.
7. Recover INCOME.XLS using multiple forensic recovery techniques, including icat, blkcat, and tsk_recover`.
8. Validate the recovered evidence using MD5 and SHA-256 cryptographic hashes and compare independently recovered copies.
9. Examine the recovered spreadsheet locally and verify its file type and structure without modifying the original forensic image.

10. Preserve command outputs, screenshots, recovered evidence, and investigative notes to create a repeatable forensic record.

11. Produce a professional forensic report documenting the methodology, evidence examined, findings, limitations, and conclusion.

The overall objective is to demonstrate that digital evidence can be identified, recovered, examined, validated, and documented in a systematic and forensically defensible manner while preserving the integrity of the original evidence.

Executive Summary

This report documents the forensic examination of the authorized training disk image Ch01InChap01.dd, conducted as part of the ICDFA digital forensics laboratory exercise. The examination was performed within a controlled Kali Linux virtual environment using Autopsy and The Sleuth Kit.

The primary objectives were to preserve and verify the integrity of the forensic image, identify deleted files, locate and recover the relevant INCOME.XLS spreadsheet, perform keyword searches, examine filesystem metadata and data units, validate recovered evidence using cryptographic hashes, and maintain a repeatable record of all significant investigative activities.

The examination identified four deleted files: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. A recursive filesystem analysis also identified INCOME.XLS at metadata address 13. istat examination showed that the file was allocated, had a size of 13,824 bytes, and occupied sectors 285–311.

The spreadsheet was recovered using icat and blkcat, with the recovered file size matching the metadata-recorded size. File-format examination identified the recovered file as a Composite Document File V2 associated with Microsoft Excel. The file header and extracted strings provided additional evidence that the recovered file was a valid legacy Excel.XLS document.

Cryptographic validation showed that the icat recovery and the existing tsk_recover copy produced identical MD5 and SHA-256 hashes. The matching hashes provide strong evidence that the two independently recovered copies contain identical data.

Keyword analysis also identified an ASCII occurrence of INCOME within the forensic image, while searches for the Unicode representation and the complete filename income.xls produced no corresponding hits.

Throughout the examination, recovered evidence and investigative outputs were maintained separately from the original forensic image. The original evidence image was not intentionally modified during the investigation.

The examination demonstrates the successful application of fundamental digital forensic techniques, including evidence preservation, filesystem analysis, deleted-file identification, metadata examination, forensic recovery, keyword searching, cryptographic validation, and documentation. The results provide a repeatable technical record of the forensic examination and support the findings presented in this report.

Tools Used

The forensic examination was conducted using Kali Linux running in a VMware Workstation virtual environment. Autopsy 2.24 was used for forensic case management, filesystem analysis, deleted-file identification, keyword searching, evidence examination, tagging, and reporting.

The Sleuth Kit (TSK) command-line tools were used extensively throughout the investigation. The `img_stat` utility was used to examine the forensic image, while `mmls` was used to analyze the partition structure and identify filesystem boundaries. `fsstat` was used to examine filesystem information, and `fls` was used to list files and directories, including deleted entries. The `istat` utility was used to examine the metadata associated with **INCOME.XLS**, including its metadata address, file size, timestamps, allocation status, and data sectors.

The `icat` utility was used to recover **INCOME.XLS** using its metadata address, while `blkcat` was used to recover the file directly from its allocated data sectors. `tsk_recover` was also used to perform filesystem-based file recovery.

For evidence validation, `md5sum` and `sha256sum` were used to calculate cryptographic hashes and compare independently recovered copies of the spreadsheet. The `file` utility was used to identify the recovered file type, while `strings` were used to extract readable content from the recovered spreadsheet. `Xxd` was used to examine the hexadecimal file header and verify the document signature. In addition, `grep` was used to filter and locate relevant filenames and keyword-search results, while `tee` was used to save important command outputs for documentation and repeatability.

Set up the working folder structure and calculate baseline cryptographic hashes for all files to ensure nothing is changed during the analysis.

1. Dedicated lab folder structure.

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ cp ~/Downloads/Ch01InChap01.dd ~/forensics/ICDFA_Lab1/evidence

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Aug 27 08:39 Ch01InChap01.dd
```

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ cd ..

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ tree
.
├── evidence
│   └── Ch01InChap01.dd
├── output
│   ├── Ch01InChap01.dd.md5
│   └── Ch01InChap01.dd.sha256
├── recovered
└── screenshots

5 directories, 3 files

```

2. Evidence image filename, size and storage location.

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ ls -lh /home/kali/Downloads/Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Aug 26 11:33 /home/kali/Downloads/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ stat /home/kali/Downloads/Ch01InChap01.dd
  File: /home/kali/Downloads/Ch01InChap01.dd
  Size: 1474560      Blocks: 2880      IO Block: 4096   regular file
Device: 8,1      Inode: 3561152      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-08-27 17:29:24.978959696 -0400
Modify: 2026-08-26 11:33:53.860354796 -0400
Change: 2026-08-26 11:33:53.892374437 -0400
 Birth: 2026-08-26 11:33:39.483221965 -0400

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ █

```

3. MD5 and SHA-256 hashes of Ch01InChap01.dd.

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Aug 27 08:39 Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ md5sum Ch01InChap01.dd > ../output/Ch01InChap01.dd.md5
sha256sum Ch01InChap01.dd > ../output/Ch01InChap01.dd.sha256

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ cat ../output/Ch01InChap01.dd.md5
cat ../output/Ch01InChap01.dd.sha256
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$

```

4. Completed mini chain-of-custody/evidence worksheet.

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ df -h ~
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       79G   44G   31G   59% /

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ cp --preserve=all /home/kali/Downloads/Ch01InChap01.dd ~/forensics/ICDFA_Lab1/evidence/

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ ls -lh ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Aug 26 11:33 /home/kali/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ md5sum ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbbcb /home/kali/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ sha256sum ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 /home/kali/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1]

```

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ cat ~/forensics/ICDFA_Lab1/report/chain_of_custody.txt
ICDFA DIGITAL FORENSICS LAB
MINI CHAIN-OF-CUSTODY / EVIDENCE WORKSHEET

Case Name:
Ch01InChap01_Investigation

Host:
Ch01InChap01

Evidence ID:
E-001

Evidence Description:
Ch01InChap01.dd - authorised training forensic disk image

Original Evidence:
Ch01InChap01.dd

Evidence Location:
/home/kali/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

Evidence Type:
Raw forensic disk image

Purpose:
Identification and recovery of deleted files and relevant Office
documents, keyword searching, metadata analysis, integrity
verification, and forensic reporting.

Original Evidence Modified:
NO

```

```

Analysis Performed On:
Working forensic image

MD5:
[a117773bcf1fc88ec0ab8e0a349fbbcb]

SHA-256:
[3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122]

Tools:
Autopsy 2.24
The Sleuth Kit
img_stat
nmmls
fsstat
fls
istat
icat
blkcat
tsk_recover
md5sum
sha256sum

Evidence Handling Statement:
The original forensic image was preserved and not intentionally
modified during examination. Recovered files and analysis outputs
were stored separately from the source image.

Examiner:
[Afeez Bolaji]

Date:
[28-8-2026]

Signature:

```

5. Autopsy case creation and evidence source configuration.

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ IMG=~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ echo "$IMG"
/home/kali/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ display

```



6. Autopsy deleted-file listing.

fls

Identify deleted files (* designation) and extract their assigned inode/metadata addresses.

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ fls -r -d Ch01InChap01.dd
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$
```

The forensic image Ch01InChap01.dd was examined using The Sleuth Kit fls utility with recursive and deleted-file options (fls -r -d). Four deleted files were identified: Billing Letter.doc (inode 8), confirmation.txt (inode 11), letter1.txt (inode 15), and Regrets.doc (inode 17).

7. Autopsy keyword-search results.

```
(kali㉿kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/keyword_results.txt
ThomasGeorge567 Port AveSeattleWA98118+8$
George Montgomery
  HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
George
Thomas George
George
  HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
```

```
(kali㉿kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/output/keyword_search_summary.txt
ICDFA LAB - KEYWORD SEARCH RESULTS

Keyword: INCOME
Search type: ASCII
Case: Case Sensitive
Result: 1 occurrence

Location:
Unit 19 (Hex - ASCII)
1: 320 (INCOME XLS)

Unicode search:
0 hits

Keyword: income.xls
Search type: ASCII
Case: Case Insensitive
Result: 0 hits
```

8. Tagged/recovered Autopsy evidence and generated report.

```
(kali㉿kali)-[~]  
$ find ~/forensics/ICDFA_Lab1/recovered -maxdepth 1 -type f -printf '%f\n'  
Regrets.doc  
Billing Letter.doc  
letter1.txt  
confirmation.txt
```

9. img_stat, mmls/fsstat and fls outputs.

img_stat & mmls

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]  
$ img_stat Ch01InChap01.dd  
IMAGE FILE INFORMATION  
-----  
Image Type: raw  
  
Size in bytes: 1474560  
Sector size: 512  
  
(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]  
$ mmls Ch01InChap01.dd  
  
(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]  
$ ls -lh Ch01InChap01.dd  
-rw-rw-r-- 1 kali kali 1.5M Aug 27 08:39 Ch01InChap01.dd
```

fsstat

Extract file system metadata, cluster sizes, FAT table structures, and volume labels.


```

(kali@kali)-[~/forensics/ICDFA_Lab1/evidence]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3M

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
-----
33-236 (204) -> EOF
285-311 (27) -> EOF

```

Gather FAT12 file system layout details

FIs output

```

(kali@kali)-[~/forensics/ICDFA_Lab1/evidence]
$ fls -r -d Ch01InChap01.dd
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali@kali)-[~/forensics/ICDFA_Lab1/evidence]
$

```

10. Recursive deleted-file listing using fls.

```

(kali@kali)-[~]
$ fls -r -d ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd | tee ~/forensics/ICDFA_Lab1/output/fls_deleted_recursive.txt
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali@kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/output/fls_deleted_recursive.txt
r/r * 8:      Billing Letter.doc
r/r * 11:     confirmation.txt
r/r * 15:     letter1.txt
r/r * 17:     Regrets.doc

(kali@kali)-[~]
$

```

11. Identification of the metadata address for INCOME.XLS.

```

(kali@kali)-[~]
$ fls -r ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd | grep -iE 'income|xls'
r/r 13: Income.xls

(kali@kali)-[~]
$ fls -r -a ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd | grep -iE 'income|xls'
r/r 13: Income.xls

(kali@kali)-[~]
$ fls -r ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd | grep -iE 'income|xls' | tee ~/forensics/ICDFA_Lab1/output/income_metadata.txt
r/r 13: Income.xls

(kali@kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/output/income_metadata.txt
r/r 13: Income.xls

```

A recursive file-system listing using The Sleuth Kit fls identified Income.xls at metadata address 13. The entry was displayed as r/r 13: Income.xls and did not contain the deleted-file indicator (*).

12. INCOME.XLS recovery using icat.

```

(kali@kali)-[~]
$ icat ~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd 13 > ~/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS

(kali@kali)-[~]
$ ls -lh ~/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS
-rw-rw-r-- 1 kali kali 14K Aug 28 10:35 /home/kali/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS

(kali@kali)-[~]
$ file ~/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS
/home/kali/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0

```

13. istat output showing file size and relevant data units.

istat Ch01InChap01.dd 8

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ istat Ch01InChap01.dd 8
Directory Entry: 8
Not Allocated
File Attributes: File, Archive
Size: 24064
Name: _ILLIN~1.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:28 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:05 (EST)

Sectors:
237 238 239 240 241 242 243 244
245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260
261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276
277 278 279 280 281 282 283

```

istat analysis of directory entry 8 showed that the file was not allocated, confirming that it was deleted. The file size was 24,064 bytes and its short filename was _ILLIN~1.DOC. The file was written on 9 December 2005 at 06:50:28 EST, accessed at 00:00:00 EST, and created at 06:59:05 EST. The file occupied sectors 237 through 284 of the forensic image.

istat Ch01InChap01.dd 11

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ istat Ch01InChap01.dd 11
Directory Entry: 11
Not Allocated
File Attributes: File, Archive
Size: 227
Name: _ONFIR~1.TXT

Directory Entry Times:
Written:      2005-12-09 06:52:58 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:06 (EST)

Sectors:
284

```

istat analysis of directory entry 11 showed that _ONFIR~1.TXT was not allocated, confirming that the file was deleted. The file size was 227 bytes. Its recorded timestamps were Written: 9 December 2005 06:52:58 EST, Accessed: 9 December 2005 00:00:00 EST, and Created: 9 December 2005 06:59:06 EST. The file's data was located in sector 284.

istat Ch01InChap01.dd 15

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ istat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:09 (EST)

Sectors:
312

```

istat analysis of directory entry 15 showed that _ETTER1.TXT was not allocated, confirming that the file was deleted. The file size was 121 bytes. Its recorded timestamps were Written: 9 December 2005 06:51:50 EST, Accessed: 9 December 2005 00:00:00 EST, and Created: 9 December 2005 06:59:09 EST. The file's data was located in sector 312.

istat Ch01InChap01.dd 17

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ istat Ch01InChap01.dd 17
Directory Entry: 17
Not Allocated
File Attributes: File, Archive
Size: 23552
Name: _EGRETS.DOC

Directory Entry Times:
Written:      2005-12-09 06:50:52 (EST)
Accessed:     2005-12-09 00:00:00 (EST)
Created:      2005-12-09 06:59:09 (EST)

Sectors:
313 314 315 316 317 318 319 320
321 322 323 324 325 326 327 328
329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344
345 346 347 348 349 350 351 352
353 354 355 356 357 358

```

istat analysis of directory entry 17 showed that _EGRETS.DOC was not allocated, confirming that the file was deleted. The file size was 23,552 bytes. Its recorded timestamps were Written: 9 December 2005 06:50:52 EST, Accessed: 9 December 2005 00:00:00 EST, and Created: 9 December 2005 06:59:09 EST. The file's data was stored in sectors 313 through 358.

14. blkcat recovery attempt and resulting evidence.

```

(kali@kali)-[~/forensics/ICDFA_Lab1]
$ fsstat evidence/Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
OEM Name: .k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3M
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848
FAT CONTENTS (in sectors)
-----
33-236 (204) -> EOF
285-311 (27) -> EOF

```

```

(kali@kali)-[~/forensics/ICDFA_Lab1]
$ blkcat -f fat12 evidence/Ch01InChap01.dd 284
Laura,
The domain has been registered and you can upload your
files by going to this ftp address using your browser.

ftp.acmeserver.com

login id: laura.roper
password: 342rroi9

Thank you for your business

George

(kali@kali)-[~/forensics/ICDFA_Lab1]
$ blkcat -f fat12 evidence/Ch01InChap01.dd 312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George

```

The fsstat utility confirmed that the forensic image uses the FAT12 file system with a sector size and cluster size of 512 bytes. Based on the data-unit information obtained from istat, blkcat was used to examine the data associated with deleted files. Sector 284 was examined for confirmation.txt, while sector 312 was examined for letter1.txt.

15. Hash comparison of recovered INCOME.XLS copies, including tsx_recover where applicable.

```
(kali@kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/output/income_hash_comparison.txt
=== INCOME.XLS RECOVERY HASH COMPARISON ===

MD5:
6a2e65afc5af4fc5f9da2859df134eac /home/kali/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS
6a2e65afc5af4fc5f9da2859df134eac /home/kali/forensics/ICDFA_Lab1/recovered/INCOME_blkcat.XLS
6a2e65afc5af4fc5f9da2859df134eac /home/kali/forensics/ICDFA_Lab1/tsx_recovered/Income.xls

SHA-256:
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/kali/forensics/ICDFA_Lab1/recovered/INCOM
E_icat.XLS
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/kali/forensics/ICDFA_Lab1/recovered/INCOM
E_blkcat.XLS
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d /home/kali/forensics/ICDFA_Lab1/tsx_recovered/I
ncome.xls
```

16. Evidence that the recovered spreadsheet was successfully examined or converted locally.

```

(kali@kali)-[~]
$ cat ~/forensics/ICDFA_Lab1/output/income_local_examination.txt
=== LOCAL EXAMINATION OF RECOVERED INCOME.XLS ===

--- File Type ---
/home/kali/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS: Composite Document File V2 Document, Little Endian, Os
: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Crea
ng Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:
52:18 2005, Security: 0

--- File Size ---
-rw-rw-r-- 1 kali kali 14K Aug 28 10:35 /home/kali/forensics/ICDFA_Lab1/recovered/INCOME_icat.XLS

--- XLS HEADER ---
00000000: d0cf 11e0 a1b1 1ae1 0000 0000 0000 0000 .....

--- RECOGNIZABLE CONTENT ---
Amelia Phillips
"$#",##0_);\_("$#",##0\
"$#",##0_);[Red]\("$#",##0\
"$#",##0.00_);\("$#",##0.00\
"$#",##0.00_);[Red]\("$#",##0.00\
\_("$"* #,##0_);\_("$"* \(#,##0\);\_("$"* "-");\_(@_)
\_(* #,##0_);\_(* \(#,##0\);\_(* "-");\_(@_)
\_("$"* #,##0.00_);\_("$"* \(#,##0.00\);\_("$"* "-"?");\_(@_)
\_(* #,##0.00_);\_(* \(#,##0.00\);\_(* "-"?");\_(@_)

```

```

--- RECOGNIZABLE CONTENT ---
Amelia Phillips
"$#",##0_);\_("$#",##0\
"$#",##0_);[Red]\("$#",##0\
"$#",##0.00_);\_("$#",##0.00\
"$#",##0.00_);[Red]\("$#",##0.00\
\_("$"* #,##0_);\_("$"* \(#,##0\);\_("$"* "-");\_(@_)
\_(* #,##0_);\_(* \(#,##0\);\_(* "-");\_(@_)
\_("$"* #,##0.00_);\_("$"* \(#,##0.00\);\_("$"* "-"?");\_(@_)
\_(* #,##0.00_);\_(* \(#,##0.00\);\_(* "-"?");\_(@_)
January
February
January Cash Flow
Income
Laura Roper
Randall Watson
Earnest Bell
Frank Haron
Thomas George
Contact
Confirmation
Grand Total
HP DeskJet 820C Series Printer
Amelia Phillips
Amelia Phillips
Microsoft Excel
Starships CMS
January
February
Worksheets

```

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ mkdir -p ~/forensics/ICDFA_Lab1/recovered

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ icat Ch01InChap01.dd 8 > ~/forensics/ICDFA_Lab1/recovered/"Billing Letter.doc"

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ icat Ch01InChap01.dd 11 > ~/forensics/ICDFA_Lab1/recovered/confirmation.txt

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ icat Ch01InChap01.dd 15 > ~/forensics/ICDFA_Lab1/recovered/letter1.txt

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ icat Ch01InChap01.dd 17 > ~/forensics/ICDFA_Lab1/recovered/"Regrets.doc"

(kali㉿kali)-[~/forensics/ICDFA_Lab1/evidence]
$ ls -lh ~/forensics/ICDFA_Lab1/recovered/
total 56K
-rw-rw-r-- 1 kali kali 24K Aug 27 11:59 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 227 Aug 27 12:00 confirmation.txt
-rw-rw-r-- 1 kali kali 121 Aug 27 12:00 letter1.txt
-rw-rw-r-- 1 kali kali 23K Aug 27 12:01 Regrets.doc

```

Tsk_recover

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ tsk_recover -e evidence/Ch01InChap01.dd ~/forensics/ICDFA_Lab1/tsk_recovered/
Files Recovered: 6

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ ls -lh ~/forensics/ICDFA_Lab1/tsk_recovered/
total 176K
-rw-rw-r-- 1 kali kali 24K Aug 27 12:58 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 102K Aug 27 12:58 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 227 Aug 27 12:58 confirmation.txt
-rw-rw-r-- 1 kali kali 14K Aug 27 12:58 Income.xls
-rw-rw-r-- 1 kali kali 121 Aug 27 12:58 letter1.txt
-rw-rw-r-- 1 kali kali 23K Aug 27 12:58 Regrets.doc

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ find ~/forensics/ICDFA_Lab1/tsk_recovered/ -type f
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/Regrets.doc
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/Income.xls
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/Billing Letter.doc
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/letter1.txt
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/confirmation.txt
/home/kali/forensics/ICDFA_Lab1/tsk_recovered/Client Info.mdb

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ ls -lh ~/forensics/ICDFA_Lab1/tsk_recovered/
total 176K
-rw-rw-r-- 1 kali kali 24K Aug 27 12:58 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 102K Aug 27 12:58 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 227 Aug 27 12:58 confirmation.txt
-rw-rw-r-- 1 kali kali 14K Aug 27 12:58 Income.xls
-rw-rw-r-- 1 kali kali 121 Aug 27 12:58 letter1.txt
-rw-rw-r-- 1 kali kali 23K Aug 27 12:58 Regrets.doc

```

The tsk_recover utility was used to recover files from the forensic image. The recovery operation successfully produced six files: Billing Letter.doc (24 KB), Client Info.mdb (102 KB), confirmation.txt (227

bytes), Income.xls (14 KB), letter1.txt (121 bytes), and Regrets.doc (23 KB). The recovered files were stored in a separate tsk_recovered directory, leaving the original evidence image unchanged.

Autopsy keyword-search results.

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "george"
ThomasGeorge567 Port AveSeattleWA98118+8$
George Montgomery
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
George
Thomas George
George
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
```

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "laura"
LauraRoper48 Mockingbird LaneSeattleWA981190+)"
@1www.lauras_stuff.comGroups.com123.13.78.231INterNICE=06
Laura Roper
Dear Laura,
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
Laura,
login id: laura.roper
Laura Roper
```

```
(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "ftp"
files by going to this ftp address using your browser.
ftp.acmeserver.com

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "password"
password: 342rroi9

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "nowhere.com"
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a evidence/Ch01InChap01.dd | grep -i "lauras"
@1www.lauras_stuff.comGroups.com123.13.78.231INterNICE=06
HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
```

KEY RESULT

KEYWORDS	EVIDENCE FOUND
George	George Montgomery, Thomas George, and george.montgomery@nowhere.com

Laura	Laura Roper, Dear Laura, laura.roper, www.lauras_stuff.com
FTP	ftp.acmeserver.com and reference to uploading files
Password	Password field found in the evidence
Nowhere.com	george.montgomery@nowhere.com
Laura	www.lauras_stuff.com and hyperlink information

```

(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$ strings -a ~/forensics/ICDFA_Lab1/recovered/"Billing Letter.doc" | grep -Ei "George|Laura|Earl|ftp|http|www|@"
Laura Roper
Dear Laura,
  HYPERLINK http://www.lauras_stuff.com
www.lauras_stuff.com
George Montgomery
  HYPERLINK mailto:george.montgomery@nowhere.com
george.montgomery@nowhere.com
George Montgomery
(kali㉿kali)-[~/forensics/ICDFA_Lab1]
$

```

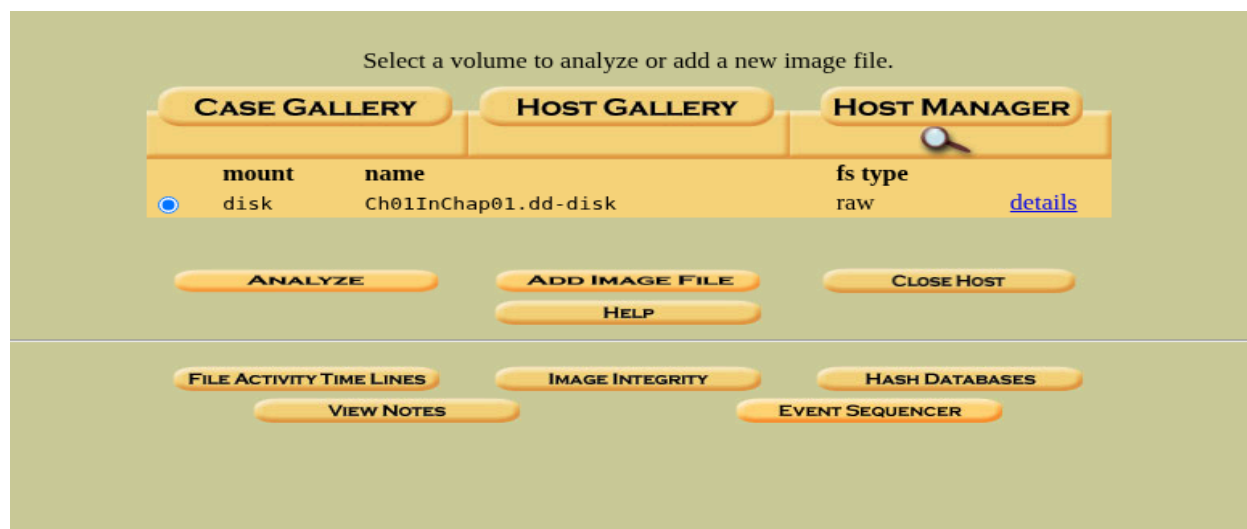
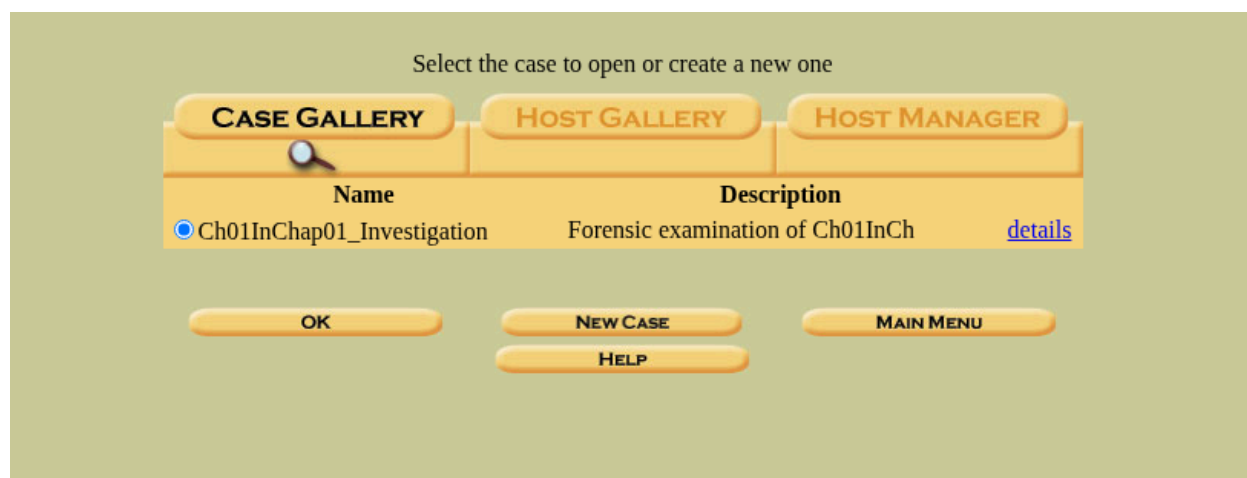
FILE	KEYWORDS/HINTS	ASSESSMENT
Billing Letter.doc	Laura Roper; George Montgomery; www.lauras_stuff.com; george.montgomery@nowhere.com	Relevant

FILE	KEYWORDS HINTS	ASSESSMENT
Regrets.doc	George Montgomery; george.montgomery@nowhere.com	Potentially relevant

The database contained client-identification/contact information without unnecessarily reproducing personal address details.

The IP addresses and website entries are also potentially useful investigative artifacts.

Part A — Autopsy Analysis



Analyze the disk

File Analysis

Overview

The File Analysis mode allows one to analyze an image from the file and directory perspective. This provides the same interface that users typically use with a normal computer. This mode will also display information about deleted files though.

This mode will allow one to examine the contents of files and directories for evidence. It even allows one to perform basic binary analysis by extracting the ASCII strings from a binary file. The files can also be sorted by any field.

Directory List

The left-hand side window has four main options:

- Directory Seek
- File Name Search
- Hide / Expand Directories
- Show All Deleted Files

By default, the directory listing is not shown when this mode is first entered. By select the [Expand Directories](#) button, the full list of directories will be shown. The number of '+' symbols represent the depth of the directory. As this is an HTML interface and there is no state management occurring (no cookies or session id), it would be difficult to have an interface where one or more directories are expanded yet the rest are collapsed.

Selecting the [All Deleted Files](#) link will display all of the deleted files in the image on the right-hand side.

There is a text box where a directory (or file) name can be entered and it will be displayed on the right-hand side. This makes it easy to jump to a directory without going through the directory listings. For example, to seek to the 'windows\system32' folder, you can enter that string into the box instead of scrolling through the directories.

There is also a text box where a pattern of a file name can be entered and all files that match that pattern will be displayed on the right-hand side. The search pattern is a Perl regular expression, so some values, such as '.' or '*' will need to be escaped. The search is done case insensitive. To find all files that have a JPG extension, the following could be used "\.jpg". Or to find all files that begin with a dot, then we could use "\.",

Directory Contents

The window in the upper right-hand side contains the directory contents. In a file system, a directory allocates data units on the disk and fills the data units with structures that contain the name of the file and the address of the meta data structure. This view parses the file name structures in the directory. It is filled by either selecting a directory from the left-hand side directory list or a directory from within the same window. The entries can be resorted by clicking on any of the header values.

The column headers have the following definitions:

- **del:** A check in this column represents a deleted file. See below for a discussion on the different color types of deleted files.
- **Type:** This column contains the "type" that the file or directory is. There are two values. The first (*dir*) is the type according to the directory entry structure. The directory entry is where the file name is located. The second value (*in*) is the type according to the meta data structure. These should be the same except for deleted files. They are both given to help the investigator identify if a deleted file has been reallocated or not. For example, if the meta data structure type is different than the file name type, then it is likely that one of the structures was reallocated for a new file.
- **Name:** The name of the file or directory. Clicking on the name displays file contents in the bottom window or directory contents in the same window. For deleted files, no link will exist if the meta data structure address has been cleared (which many OS do). To identify the destination of a UNIX symbolic link, click on the address at the end of the row. Note that the name will not be a link if the size of the file is 0 or if the meta data structure is unknown.
- **Modified Time:** This column exists for UNIX and NTFS file systems. It shows the last time that the file data was last modified. In other words, when was data last written to the data units allocated by the file. In UNIX, the `utimes()` function can modify this value very easily.
- **Written Time:** This column exists for FAT file systems and is the time when the file was last written to. Of the three times, this is the only value that is required by the FAT specification.
- **Accessed Time:** This column contains the last accessed time of the file data. On a FAT image, this value is optional and is only accurate to the day (not hours or seconds). This value can be modified by the `utimes()` function in UNIX.
- **Changed Time:** This column exists for UNIX and NTFS file systems. It is the last time that the file status (or meta data) was changed. This is different than the Modified time because modified deals with the file data and this deals with the descriptive data in the inode or MFT entry. This value cannot be changed by the `utimes()` function in UNIX.
- **Created Time:** This column exists for NTFS and FAT file systems. It is the time when the file was created. According to the FAT spec, it is an optional time.
- **Size:** The size of the file. Note that if the size of the file is 0, no link will be shown with the name.
- **UID:** The User ID of the file owner.
- **GID:** The Group ID of the file owner.
- **Meta Data:** The file name structures contain a pointer to the meta data structure that describes the file. This column contains the address of the structure. Selecting this value will display the details in the bottom window. If the value is invalid, then a link will not exist. If it is for a deleted file name that has a meta data structure with an allocated status, a "realloc" string will exist to identify this.

The Add Note link allows you to make a comment about this directory and have it saved in your personal notes file.

The [Generate MD5 List](#) link will generate the MD5 value for every file in the directory and allow you to save it as a text file. Using fingerprint data bases, This makes it easy to check for files that were modified by an attacker.

The path on top of the window has hyperlinks in it that allow the user to easily change to a previous directory.

There are two different colors used for deleted files. The difference is based on the status of the data structures in the file. A **bright red** entry means that the file name data structure is not allocated and the meta data structure that it points to is also not allocated. This is what we would expect of a recently deleted file. This means that we can trust the data we are seeing as long as the meta data structure was not allocated and unallocated since the deletion. If it is **darker red**, then the meta data structure has been reallocated and the data is most likely not accurate.

Keyword search

Searching for ASCII: Done Saving: Done 1 hits- link to results
Searching for Unicode: Done Saving: Done 0 hits
New Search
1 occurrence of INCOME was found Search Options: ASCII Case Sensitive
Unit 19 (Hex - Ascii) 1: 320 (INCOME XLS)
INCOME was not found Search Options: Unicode Case Sensitive

Searching for ASCII: Done

Saving: Done

1 hits- [link to results](#)

Searching for Unicode: Done

Saving: Done

0 hits

[New Search](#)

1 occurrence of INCOME was found

Search Options:

ASCII

Case Sensitive

Unit 19 ([Hex](#) - [Ascii](#))

1: 320 (INCOME XLS)

INCOME was not found

Search Options:

Unicode

Case Sensitive

Searching for ASCII: Done

Saving: Done

0 hits

[New Search](#)

income.xls was not found

Search Options:

ASCII

Case Insensitive

Unit Number:

Number of Units:

Unit Size: 512

Lazarus Addr: ☐

VIEW

Investigation Findings

The forensic examination of the authorized disk image ****Ch01InChap01.dd**** was conducted using Autopsy and The Sleuth Kit within a controlled Kali Linux forensic environment. The original forensic image was preserved and was not modified during the examination.

1. Deleted Files Identified

Metadata Address	File Name	Status
8	Billing Letter.doc	Deleted
11	confirmation.txt	Deleted
15	letter1.txt	Deleted
17	Regrets.doc	Deleted

The entries were displayed with the ```` deleted-file indicator in the ``ls`` output.

2. Identification of INCOME.XLS

A recursive ``ls`` examination identified the file:
INCOME.XLS

at metadata address 13
``text``
r/r 13: Income.xls

Unlike the four entries listed above, this entry did not contain the deleted-file indicator and was therefore identified by the filesystem as an allocated file.

3. Metadata Examination of INCOME.XLS

The metadata associated with ``INCOME.XLS`` was examined using ``istat``.

The examination established:

- * Metadata address: 13
- * Allocation status: Allocated
- * File size: 13,824 bytes
- * File attributes: File, Archive
- * Written: 2005-12-09 06:52:18 EST
- * Accessed: 2005-12-09 00:00:00 EST
- * Created: 2005-12-09 06:59:07 EST
- * Data sectors: 285–311

4. Recovery of INCOME.XLS

The spreadsheet was recovered using multiple The Sleuth Kit techniques.

Using icat with metadata address 13, the file was recovered as:
recovered/INCOME_icat.XLS

The file was also recovered directly from data sectors ****285–311**** using `blkcat` and saved as:
recovered/INCOME_blkcat.XLS

The blkcat recovery produced a file size of 13,824 bytes, matching the size reported by istat.

5. Recovered File Validation

The blkcat recovered file was identified by the file utility as a: Composite Document File V2 Document with Microsoft Excel identified as the creating application.

The file header was: D0 CF 11 E0 A1 B1 1A E1

This is consistent with the OLE Compound File format used by legacy Microsoft Excel.XLS documents.

The recovered spreadsheet also contained recognizable Excel-related content, including worksheet information, months, financial labels, and names.

6. Hash Verification of Recovered Copies

The icat recovery and the existing tsk_recover copy were independently hashed.

Both copies produced the same values:

MD5 6a2e65afc5af4fc5f9da2859df134eac

SHA-256 8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

The matching MD5 and SHA-256 values demonstrate that these two recovered copies are byte-for-byte identical.

A three-way comparison including the blkcat copy should be included in the final evidence once the final hash comparison has been captured.

7. Keyword Search Finding

A keyword search for INCOME produced one ASCII hit:

Unit 19 (Hex - ASCII)

1: 320 (INCOME XLS)

The corresponding Unicode search produced zero hits.

A separate case-insensitive ASCII search for income.xls produced zero hits. This demonstrates that keyword-search results depend on the exact search term and representation present within the raw image.

Overall Finding

The examination successfully identified deleted files, located the INCOME.XLS filesystem entry, examined its metadata and allocated data sectors, recovered the spreadsheet using icat and blkcat, and validated the recovered document through file-format and cryptographic-hash examination.

The evidence supports the conclusion that INCOME.XLS is a valid legacy Microsoft Excel document present at metadata address 13, with its data allocated across sectors 285–311. The icat and tsk_recover copies have matching MD5 and SHA-256 values, providing strong evidence that the two recovery methods produced identical file content.

Technical Report

1. Scope and Methodology

The examination was conducted on the authorized forensic disk image Ch01InChap01.dd as part of the IC DFA digital forensics laboratory exercise. The objective was to identify deleted files, locate and recover the relevant INCOME.XLS spreadsheet, perform keyword searching, examine filesystem metadata and data units, validate recovered evidence using cryptographic hashes, and preserve a repeatable record of the investigation.

The examination was performed in a Kali Linux virtual machine. The original forensic image was maintained separately from recovered evidence and working files and was not intentionally modified during the examination.

The investigation used Autopsy and The Sleuth Kit, including:

- img_stat — forensic image information
- mmls — partition/layout analysis
- fsstat — filesystem information
- fls — file and directory listing
- istat — metadata examination
- icat — file recovery using metadata address
- blkcat — data-unit recovery
- tsk_recover — filesystem-based file recovery
- md5sum — MD5 integrity verification
- sha256sum — SHA-256 integrity verification
- file — file-format identification
- strings — extraction of readable strings
- xxd — hexadecimal header examination

2. Evidence Examination

The forensic image was stored in the dedicated evidence directory:

~/forensics/ICDFA_Lab1/evidence/Ch01InChap01.dd

The investigation maintained separate directories for evidence, recovered files, command outputs, screenshots, and TSK recovery results.

The deleted-file examination identified four deleted entries:

8: Billing Letter.doc
11: confirmation.txt
15: letter1.txt
17: Regrets.doc

The recursive filesystem examination also identified INCOME.XLS at metadata address 13.

3. INCOME.XLS Metadata Examination

istat was used against metadata address 13. The file was reported as allocated and had a size of 13,824 bytes.

The metadata examination identified the following data sectors: 285–311

These 27 sectors correspond to the reported file size when calculated using the image's 512-byte sector size.

The recorded timestamps were:

Written: 2005-12-09 06:52:18 EST
Accessed: 2005-12-09 00:00:00 EST
Created: 2005-12-09 06:59:07 EST

4. File Recovery

INCOME.XLS was recovered using metadata address 13 with icat and saved separately as:
recovered/INCOME_icat.XLS

The file was also recovered from sectors 285–311 using blkcat and saved as:
recovered/INCOME_blkcat.XLS

The blkcat recovery produced exactly 13,824 bytes, consistent with the size reported by istat.

A separate copy was available from the tsk_recover process:
tsk_recovered/Income.xls

5. File Validation

The recovered spreadsheet was examined locally using file, strings, and xxd.

The file utility identified the recovered blkcat output as a Composite Document File V2 Document, with Microsoft Excel identified as the creating application.

The hexadecimal header began with:

D0 CF 11 E0 A1 B1 1A E1

which is consistent with the OLE Compound File format used by legacy Microsoft Excel .XLS documents.

The strings examination recovered recognizable spreadsheet content, including Microsoft Excel, January, February, Income, Grand Total, Worksheets, and multiple names.

6. Cryptographic Validation

The icat recovery and the tsf_recover copy produced identical cryptographic hashes:

MD5:

6a2e65afc5af4fc5f9da2859df134eac

SHA-256:

8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d

The matching values demonstrate that the two recovered copies are byte-for-byte identical.

The final three-way hash comparison involving the blkcat copy should be retained as supporting evidence if required by the assessment.

7. Keyword Examination

Keyword searching was performed against the raw forensic image.

The ASCII search for:

INCOME

produced one hit:

Unit 19 (Hex - ASCII)

1: 320 (INCOME XLS)

The Unicode search produced zero hits.

A separate case-insensitive ASCII search for: income.xls

produced zero hits.

These results were preserved as part of the investigation evidence.

8. Evidence Preservation

Recovered files were stored outside the original evidence-image directory. Command outputs were saved in the investigation's output directory, while screenshots were maintained separately.

The original Ch01InChap01.dd image was used as the source for analysis and recovery and was not intentionally modified.

All significant commands, findings, recovery activities, and validation results were recorded to support repeatability of the examination.

Limitations

The following limitations were identified during the examination:

1. The inves

tigation was performed in a controlled laboratory environment using an authorized training forensic image. Therefore, the findings should not automatically be generalized to a live criminal or corporate investigation.

2. The original evidence image was examined using available forensic tools within the Kali Linux virtual machine. Differences in tool versions, filesystem interpretation, or operating-system configuration may produce minor differences in presentation.

3. Autopsy's keyword-search interface produced an ASCII hit for INCOME, while the separate search for the complete filename income.xls produced no hit. Keyword-search results are dependent on the exact character representation and location of the searched data.

4. The presence of readable strings within a recovered document does not by itself establish the circumstances in which the document was created, modified, or used.

5. The timestamps reported by istat represent filesystem metadata values from the forensic image. They should not independently be interpreted as proof of a person's actions or the exact real-world time of an event.

6. LibreOffice was not installed in the Kali environment, so the recovered spreadsheet was not opened in a graphical spreadsheet application. Instead, its format, header, size, and readable strings were examined using command-line forensic utilities.

7. The icat and tsk_recover copies of INCOME.XLS were confirmed to have identical MD5 and SHA-256 values. A complete three-way cryptographic comparison including the blkcat copy should be retained in the final evidence package if required for full validation.

8. The examination was limited to the contents and structures available within the supplied forensic image. No conclusions were drawn about information that was not present or recoverable from the image.

9. The findings should therefore be interpreted together with the original forensic image, command outputs, screenshots, hash records, and other supporting evidence submitted with the laboratory assessment.

Conclusion

The forensic examination of Ch01InChap01.dd successfully demonstrated the application of a structured digital forensic investigation process.

The examination identified four deleted files: Billing Letter.doc, confirmation.txt, letter1.txt, and Regrets.doc. The investigation also located INCOME.XLS at metadata address 13 and established that the file had a size of 13,824 bytes with data allocated across sectors 285–311.

The spreadsheet was successfully recovered using icat and blkcat, and the recovered data was consistent with a legacy Microsoft Excel.XLS document. File-format examination, hexadecimal header analysis, and readable-string extraction provided additional evidence supporting the identification of the recovered document.

Cryptographic verification showed that the icat recovery and the tsk_recover copy produced identical MD5 and SHA-256 values, demonstrating that those two recovery methods produced identical file content.

The investigation also demonstrated the use of keyword searching, filesystem metadata analysis, data-unit examination, file recovery, hashing, and evidence preservation. The original forensic image remained separate from recovered evidence and was not intentionally modified.

Overall, the laboratory objectives were substantially achieved, and the documented methodology provides a repeatable technical record of the examination. The accompanying screenshots, command outputs, recovered files, hash records, and forensic report provide supporting evidence for the findings presented in this report.